

Kristian Haugen

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SUMMARY

Aerospace and Mechanical Engineer with experience in liquid rocket propulsion, thermal-fluid systems, and supersonic flight testing. Background includes NASA internships, liquid propulsion system design, CFD and thermal analysis, and test infrastructure development. Proven ability to lead multidisciplinary engineering teams and deliver flight-tested hardware for high-performance aerospace applications.

EDUCATION

Bachelor of Science | *University of North Dakota, Grand Forks, ND* **May 2026**

Double Major: Aerospace Engineering & Mechanical Engineering | Double Minor: Physics & Mathematics

GPA: 3.67/4.0

Relevant Coursework: Orbital Mechanics, Space Propulsion Systems, Aerodynamics, Computational Fluid Dynamics, Spacecraft Systems, Air Breathing Engines, Thermodynamics, Flight Dynamics & Control

WORK EXPERIENCE

Aeronautics Intern, *Vorbeck Materials, Grand Forks, ND* **Jun 2025 – Present**

- Conducting technical analysis and systems-level demonstrations of high-altitude radio repeater targeting 5000 ft initial altitude supporting extended range hypersonic vehicle command and control.
- Integrating conformal graphene-based antenna arrays on sounding rocket airframes, enabling ~99% reduction in frontal area versus equivalent monopole antennas.
- Designing and validating experimental flexible graphene antenna integration methods for ground and air applications evaluating RF performance at 1.6 GHz under highly dynamic environments.

Rocket Lab Research Assistant, *University of North Dakota, Grand Forks, ND* **Mar 2024 – Aug 2025**

- Procured and commissioned \$300K+ in propulsion lab equipment, establishing infrastructure for concurrent research programs.
- Coordinated university-wide interdisciplinary collaboration on rocket projects, increasing student participation by 40%.
- Designed and implemented lab protocols and infrastructure improvements, enabling 4 concurrent student-led propulsion research projects annually.

NDSGC Student Research Fellowship, *University of North Dakota, Grand Forks, ND* **Sep 2024 – Dec 2024**

- Conducted a comprehensive literature review on rotating detonation engines, synthesizing findings to identify industry and academic research gaps.
- Authored a detailed, budget-conscious guide for undergraduate teams to design, construct, and test small-scale rotating detonation engines, enhancing accessibility to advanced propulsion studies.
- Developed a preliminary design for a compact rotating detonation engine and its ground support systems, focusing on cost-effective materials and scalability.

NASA Intern, *Glenn Research Center, Cleveland, OH* **Aug 2023 – Dec 2023**

- Performed thermal analysis and CFD simulations of orbital cryogenic fluid transfer using Thermal Desktop achieving model validation within 5% of experimental datasets.
- Briefed thermal model validation results to the GRC thermal fluids management team supporting integration into orbital fuel transfer missions.

NASA Intern, *Langley Research Center, Hampton, VA* **Jun 2023 – Aug 2023**

- Synthesized a compilation of 41 peer-reviewed sources on perovskite semiconductor properties identifying material characteristics critical to improving radiation-detection efficiency for space-based sensor applications.
- Investigated atomic oxygen degradation mechanisms on crystalline solar cells, proposed a potential exposure test to quantify crystal degradation over time.

PROJECTS

Collegiate Propulsive Lander Challenge, *UND Advanced Rocketry Club*

Sep 2024 – May 2026

- Conducted requirements analysis and preliminary design of a Methalox liquid propulsion system targeting 1,000 lbf thrust, including propellant handling systems and ground support infrastructure.
- Designed and commissioned ground support infrastructure for liquid rocket engine testing including propellant loading systems, data acquisition on 42 channels, pressure/flow instrumentation, and range safety protocols compliant with university SOPs.
- Managed systems engineering lifecycle and milestone schedule for a 15+ person multidisciplinary team, maintaining compliance with competition requirements and university safety SOPs across 4 concurrent workstreams.

International Rocket Engineering Competition 2025, *UND Advanced Rocketry Club* Oct 2024 – Jun 2025

- Led cross-functional team of 20+ engineers in the design, manufacturing, and flight test of a high-performance rocket achieving apogee higher than 21,000 feet and reaching Mach 1.2 supersonic flight.
- Directed structural analysis and composite materials fabrication of a carbon fiber aft section, reducing aerodynamic drag for sustained Mach 1.2 supersonic flight.
- Coordinated full project lifecycle, systems integration, and flight test validation of all subsystems.

International Rocket Engineering Competition 2024, *UND Advanced Rocketry Club* Oct 2023 – Jun 2024

- Applied systems engineering principles and FEA-informed structural design to lead the Airframe team in developing a rocket achieving 30,000 ft altitude at Mach 1.87.
- Designed and fabricated a custom instrumented nosecone integrating pitot sensors and thermocouple arrays to acquire airspeed and surface heating data across transonic and supersonic flight regimes up to Mach 1.87.
- Designed a selective radiation shield using bismuth attenuation layers to reject terrestrial background radiation while maintaining visibility of space-based X-rays to an onboard Radiacode detector.

SKILLS

CAD/CAE: Creo Parametric, SolidWorks, Onshape, ANSYS, Mastercam

Analysis & Programming: CFD, FEA, Thermal Desktop, Python, MATLAB, Fortran, Structural Analysis, Failure Analysis, Test Data Reduction

Manufacturing: Additive Manufacturing, CNC Machining, Metal Lathe, Composite Layup, Soldering

Propulsion: Liquid Propulsion Systems, Combustion, Propellant Handling, Methalox Propellants, Ground Support Equipment (GSE), Data Acquisition (DAQ), Pressure Fed Systems, Oxygen Compatibility and Cleaning

Soft Skills: Project Management, Technical Documentation, Risk Management, Systems Integration, Range Safety

LEADERSHIP EXPERIENCE

President, *UND Advanced Rocketry Club*

May 2023 – May 2026

- Directed operations of a 50+ member technical organization, leading ten officers across engineering, safety, logistics, and outreach functions to execute two national competitions annually.
- Negotiated and secured 1350 sq ft of dedicated lab and propulsion testing space through collaboration with university administration and faculty advisors, establishing safety compliance frameworks for high-powered rocketry operations.
- Secured over \$25,000 in sponsorships and in-kind donations by building relationships with local and national aerospace companies, enhancing club resources for advanced projects.

HONORS & AFFILIATIONS

AIAA – UND Branch President

Dec 2024 – May 2026

Tripoli Rocket Association – Member, Level 1 Certified

May 2023

Tau Beta Pi Engineering Honor Society – Member

Mar 2023

American Radio Relay League – Technician License

Jan 2023

National Association of Rocketry – Member, Level 1 Certified

May 2022